Harrison B. Tracy

201 Fincastle Drive | Blacksburg, Virginia 24060 | harrisontracy123@vt.edu | +1 (540)-553-4260

Security Clearance: DoD Secret

Education

Virginia Tech — **Blacksburg, Virginia**

M.S. Aerospace Engineering – Hypersonic Aerothermodynamics — Incoming August 2027

GPA: –/4.0

B.S. Aerospace and Ocean Engineering — August 2023 - Present

GPA: 3.94/4.0

Relevant Coursework: Applied CFD, Compressible Aerodynamics, Configuration Aerodynamics, Thermodynamics and Propulsion, Aerospace Structures, Computational Methods, Aircraft Stability and Control, Air Vehicle Design, Astromechanics

Skills

Software & Tools: StarCCM+, NX, Teamcenter, Solidworks, Altair Hyperworks (HyperMesh, OptiStruct), Abaqus

Programming Languages: Python, Java, Matlab, Julia, C/C++, HTML

Manufacturing & Prototyping: 3D Printer, CNC, Laser Cutter, Lathe

Experience

Northrop Grumman - Mission Systems — **Sunnyvale, California**

Mechanical Engineering Intern — June 2026 - August 2026

- Developed a parametric bearing modeling tool which integrated a **Python-to-NX** pipeline system to automate 3D model generation and preliminary load calculations, reducing model creation time by over **99%**
- Rebuilt a legacy submarine tilt pad bearing assembly in **NX**, then performed operational 6-degree-of-freedom stiffness calculations by hand and validated the results using **FEA**, achieving agreement within **7%** of predicted values
- Delivered rapid prototyping support for engineering teams, designing and fabricating parts on demand via **3D printing**, **laser cutting**, and **CNC machining** to accelerate design iteration and testing

HII - Newport News Shipbuilding — **Newport News, Virginia**

Fluid Systems Mechanical Engineering Intern — May 2025 - August 2025

- Modeled various Naval valve systems in **Solidworks** then conducted mesh convergence studies using an in-house **CFD** solver to optimize simulation runtime, achieving agreement within **5%** of experimental data from previous flow tests
- Conducted a by-hand **failure mode analysis** on a CVN reactor valve component to reverify a legacy factor of safety in compliance with updated Naval valve operation standards and procedures
- Updated and revised various engineering documents such as drawings, BOMs, SOWs, and operational manuals to ensure clear communication between engineering teams, customers, and vendors

Kevin T. Crofton Department of Aerospace and Ocean Engineering — **Blacksburg, Virginia**

Hypersonic Thermals and CFD Researcher — January 2026 - Present

- Modeled and simulated hypersonic flows over various deformed surfaces in **StarCCM+** to analyze the effects of surface divot geometry on boundary layer turbulence, heat circulation structures and surface heating augmentation
- Compared the effect of numerous thermodynamic gas models across varying divot depths, identifying a **120K** discrepancy in peak surface temperature and shifts in recirculation structure attributable to real-gas thermochemical effects
- Authored an extended abstract submitted to the AIAA SciTech 2027 Forum under faculty mentorship

Publications

Tracy, H. B., and Sharma Priyadarshini, M., "Effect of Surface Defects and Thermochemistry on Laminar Hypersonic Boundary Layers," AIAA SciTech 2027 Forum, 2027 {accepted}

Projects

Neural Network Surrogate Model for Hypersonic Boundary Layer Chemistry

- Developed a **neural network** trained on **1,470** CFD solver outputs to predict equilibrium mole fractions of a 5-species atmospheric gas mixture from temperature input, enabling rapid-turnaround alternatives to full CFD for hypersonic reentry and heat transfer modeling
- Preprocessed and structured CFD-generated training data, then tuned network architecture and hyperparameters to minimize prediction error across the equilibrium composition space

NAVAIR Carrier CCA Concept Design

- Leading aerodynamic design and analysis for a fully autonomous, carrier-integrated CCA concept in collaboration with NAVAIR, while contributing to avionics systems integration
- Team scope spans concept sizing, autonomy architecture, mission planning, and cost analysis.